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CONDENSED-MATTER SPECTROSCOPY

Estimation of the Spectral Absorption of Light by Components of Human Skin

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Abstract—The spectra of the absorption coefficients of external radiation by basic chromophores of human skin, such as melanin, basic tissue, and blood (including oxy- and deoxyhemoglobin), are studied in the wavelength range of 300–1000 nm. For estimates, analytical methods of the theory of light transfer are used, which take into account the multilayer structure of a biological tissue, multiple light scattering in a medium, and multiple rereflections of radiation between layers. The calculated spectra are compared with the wavelength dependences of the absorption indices of these components available from the literature. It is shown that the spectral behaviors of the coefficients and indices of absorption strongly differ, which is related to the selectivity of the optical properties of a biological tissue. The possibilities of predicting the absorption coefficients of the skin components from the absorption coefficient of the entire skin measured under conditions of variation of its biophysical parameters (the volume concentrations of melanin and blood vessels) over a wide range are evaluated.

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INTRODUCTION

Light is frequently used to exert beneficial action on the state of living organisms. In humans, the energetic activity of cell membranes increases, the bioenergetic potential of cells rises, exchange and regenerative processes are intensified, the favorable action on the immune protective system and nerve endings takes place, and so on. All this facilitates the treatment of various diseases. Light therapy has been used since long ago; its use was revived in the second half of the 20th century, when unique light sources, i.e., lasers, were developed. Various hypotheses have been proposed based on mechanisms that produce biostimulating, therapeutic, and cosmetological actions of laser radiation on biological tissues. In this case, the absorption of light both by the entire tissue and by its individual components is one of many factors of the influence of irradiation on the human organism. For example, a suntan depends on the absorption of light by the pigment melanin. Upon the absorption of light, blood oxyhemoglobin was observed to dissociate [1, 2], which leads to the formation of deoxyhemoglobin and the release of molecular oxygen, which increases the local transfer of the latter to different parts of the organism. Photodynamic and light-oxygen effects are known [3] that give rise to the formation of singlet oxygen under the action of radiation on triplet oxygen of cells. Based on these effects, photodynamic and light-oxygen methods for therapy of oncological diseases were developed [4]. The optimality of light therapy procedures depends on the dose and spectral composition of irradiation, absorption ability of tissue components, and penetration depth of light. These questions have been considered in many papers (see, e.g., [5–8]). However, theoretical (physical, chemical, and medicobiological) and experimental studies are frequently qualitative in character and clearly cannot answer all questions that arise. Therapeutic practice relies mostly on statistically processed experimental material rather than solid scientific argumentation. This study, which is based on the modern engineering theory of radiation transfer in a scattering medium [9], yields the pattern of the absorption of light both by the entire tissue and by its individual components in a wide spectral interval, which covers wavelengths used at present in light and laser therapy [8, 10].

STRUCTURE OF THE CUTANEOUS COVERING AND OPTICAL PROPERTIES OF ITS COMPONENTS

Light from the sun, as well as from various phototherapeutic lamps, for example, Bioptron lamps [11] and low-intensity lasers, acts on a tissue through the skin cover. Clearly, the structure and biophysical and optical properties of the skin cover affect both the magnitude and the spectral selectivity of light fields that are formed within biological tissues and act on all skin chromophores. With respect to light absorption, the characteristic of the light field, which determines the efficiency of this process, is the radiation density [12]. Therefore, we initially consider the structural, biophysical, and optical properties of skin, then present the summary of results on the method of calculation of depth profiles of the radiation density. For "average" or "conventionally normal" tissues, these properties have

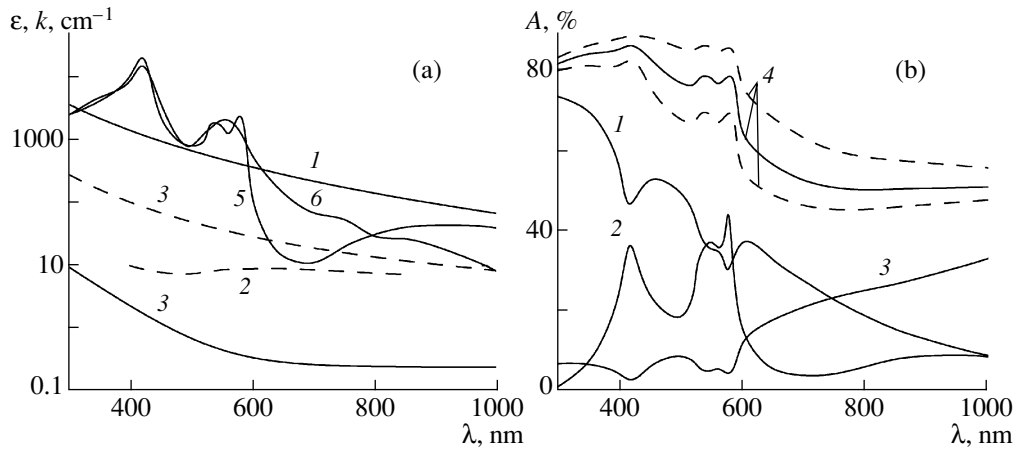

Fig. 1. Spectral dependences (a) of the (solid curves) absorption index and (dashed curves) transport extinction index of (*1*) melanin, (*2*) blood, (*3*) basic tissue, (*5*) oxyhemoglobin, and (*6*) deoxyhemoglobin and (b) of the absorption coefficient of (*1*) melanin, (*2*) blood, (*3*) basic tissue, and (*4*) entire skin.

been rather completely studied in the literature [6, 13–15]. Usually, a laminated model of skin is used whose characteristics vary along only one coordinate, i.e., depth z. Three basic layers of skin are distinguished [14], including the upper, stratum corneum, which is 0.01–0.02 mm thick; the epidermis; and the dermis. The epidermis is 0.04–0.15 mm thick, and the thickness of the dermis is in the range 1–4 mm. In turn, the dermis consists of several distinctive sublayers [13, 15, 16], which contain different amounts of blood vessels. Clearly, this structure is substantially conventional, because there are neither distinct boundaries between the layers nor sharp differences in their optical properties. According to the model of [17, 18], we will assume that the epidermis consists of a basic tissue, which absorbs and scatters light (in the literature, the basic tissue is frequently termed the bloodless tissue), and an absorbing component, melanin. The dermis consists of the same basic tissue in which blood vessels or capillaries are distributed homogeneously and randomly. Optically, this layer is assumed to be homogeneous and infinitely thick.

The optical characteristics that determine the regularities of radiation transfer in a medium are the absorption index μ_a, extinction index μ_e ($\mu_e = \mu_a + \mu_s$, where μ_s is the scattering index), and the transport extinction index $\mu_e' = \mu_e(1 - g)$ or the transport scattering index $\mu_s' = \mu_s(1 - g)$, where g is the average cosine of the scattering indicatrix. Usually, $\mu_e' \cong \mu_s'$, since, for the tissue, $\mu_a \ll \mu_s$. For the stratum corneum, epidermis, and dermis, these characteristics are denoted by the indices 1, 2, and 3, respectively, as shown below:

$$\mu_{a1} = \mu_{at}, \quad \mu_{a2} = f_m\mu_{am} + (1 - f_m)\mu_{at}, \quad \mu_{a3} = C_V\mu_{ab} + (1 - C_V)\mu_{at}, \tag{1}$$

$$\mu_{e1} = \mu_{et}, \quad \mu_{e2}' = (1 - f_m)\mu_{et}', \quad \mu_{e3}' = C_V\mu_{eb}' + (1 - C_V)\mu_{et}', \quad g_{1-3} = 0.62 + 0.00029\lambda \quad (\lambda \text{ in nm}). \tag{2}$$

Here, μ_{et}' and μ_{eb}' are the transport extinction coefficients of the basic tissue and blood; μ_{at}, μ_{am}, and μ_{ab} are the absorption indices of the basic tissue, melanin and blood; and f_m and C_V are the volume concentrations of melanin in epidermis and capillaries in dermis, respectively. In turn, the absorption index of blood is given by

$$\mu_{ab} = Hf[S\mu_{a,\,HbO_2} + (1 - S)\mu_{a,\,Hb}], \tag{3}$$

where H is the hematocrit (the volume fraction of erythrocytes in the blood), f is the volume fraction of hemoglobins in the erythrocytes, S is the oxygenation degree of the blood, and $\mu_{a,\,HbO_2}$ and $\mu_{a,\,Hb}$ are the absorption indices of oxyhemoglobin HbO_2 and deoxyhemoglobin Hb. The spectral dependences of the optical characteristics of the skin cover $\mu_{a,\,HbO_2}$, $\mu_{a,\,Hb}$, μ_{at}, μ_{am}, μ_{et}', and μ_{eb}' are presented in Fig. 1a; they were obtained as a result of critical analysis, selection, and averaging of the data from [6, 14, 15, 17–19].

REGULARITIES OF THE PROPAGATION OF LIGHT IN A TISSUE

The propagation of light in tissues has been considered in many studies. Entire issues of journals, such as *Applied Optics* [20–22] and *Journal of the Optical Society of America* [23], have been devoted to this problem. In most cases, the distribution of the absorbed energy over the depth of a tissue is determined using various numerical procedures, including the discrete ordinate method [16, 24] and the Monte Carlo method [25]. However, such calculations commonly require

large amounts of computer time; additionally, prolonged numerical calculations are sometimes published in the literature for rather simple situations, which is not justified due to the existence of reasonable analytical methods [9]. We will use the calculated structures of light fields in the skin cover [10, 26] for solving the problem posed. Since analytical expressions obtained in these studies are rather cumbersome, we will not present them here and only describe the conditions under which they were obtained.

The extinction index of skin is large. Therefore, light cannot penetrate deep into the skin and, consequently, the width of the light beam can frequently be considered to be much greater than the depth of the illuminated medium. Mathematically, this corresponds to the irradiation of a rather thick medium by an infinitely wide light beam, when the structure of the radiation field in the medium remains constant in the cross section of the beam.

The optical thicknesses of the stratum corneum and epidermis are small. Therefore, the reflection and transmission coefficients of these layers for the incident (directional) light beam are calculated in the small-angle approximation of the theory of radiation transfer [9].

The dermis is assumed to be infinitely thick and weakly absorbing. Its reflection coefficient is calculated in the asymptotic approximation of the theory of radiation transfer [9, 27].

Upon illumination of the skin surface from outside by a directional light beam, the reflection coefficient of the skin is 0.04 and, if the skin surface is illuminated from inside, this coefficient is 0.2 due to the diffusivity of light and the roughness of the surface [13, 15, 26].

The refractive index between skin layers varies smoothly so that the Fresnel reflection of light does not occur.

The reflection coefficient of a tissue and the radiation density within the medium are determined taking into account multiple rereflections of light from successive interfaces between the air, skin surface, stratum corneum, epidermis, and dermis [10, 26]. In final expressions, the radiation density is represented as the sum of geometric progressions.

Below, we will study the absorption efficiency of light by different chromophores of skin. The absorption coefficient A of the entire skin cover (or of its ith component A_i) is considered to be equal to the ratio of the light flux absorbed by the entire tissue (or by its ith component) to the flux incident to the surface of the medium. The quantities A_i were calculated as follows:

$$A = 1 - R, \tag{4}$$

$$A_{\mathrm{t}} = \mu_{\mathrm{at}}\left[\int_0^{z_1} E(z)dz + (1 - f_{\mathrm{m}})\int_{z_1}^{z_2} E(z)dz + (1 - C_{\mathrm{V}})\int_{z_2}^{\infty} E(z)dz\right], \tag{5}$$

$$A_{\mathrm{m}} = f_{\mathrm{m}}\mu_{\mathrm{am}}\int_{z_1}^{z_2} E(z)dz, \tag{6}$$

$$A_{\mathrm{HbO_2}} = C_{\mathrm{V}}HfS\mu_{\mathrm{a,\,HbO_2}}\int_{z_2}^{\infty} E(z)dz,$$

$$A_{\mathrm{Hb}} = C_{\mathrm{V}}Hf(1 - S)\mu_{\mathrm{a,\,Hb}}\int_{z_2}^{\infty} E(z)dz, \tag{7}$$

$$A_{\mathrm{b}} = A_{\mathrm{HbO_2}} + A_{\mathrm{Hb}}.$$

Here, R is the reflection coefficient of the entire skin; $0 - z_1$, $z_1 - z_2$, and $z_2 - \infty$ are the intervals of the current coordinate z in the stratum corneum, epidermis, and dermis, respectively; and $E(z)$ is the radiation density at the depth z normalized to the illumination created by the light beam on the skin surface. In what follows, we will assume that $H = 0.4$, $f = 0.25$ [15, 26], and the thickness of the epidermis is $z_2 - z_1 = 60$ μm [17]. The method of calculation of the depth profiles $E(z)$ and the corresponding calculation formulas are presented in [10].

ANALYSIS OF RESULTS

We will assume that, typically, for light human skin, $C_{\mathrm{V}} = 2\%$ and $f_{\mathrm{m}} = 8\%$ [10, 17]. For this case, Fig. 1b presents the absorption spectra of the entire skin and its components. Let us collate the latter with the spectral absorption indices shown in Fig. 1a. It is seen that there is no clearly pronounced similarity between them. The spectra strongly differ from each other. This results from the competition between absorption processes of chromophores because of their different positions in the skin, different concentrations, and absolute values of the absorption indices at one and the same wavelength of light λ. For melanin, $\mu_{\mathrm{am}}(\lambda)$ is the continuously decreasing curve, whereas $A_{\mathrm{m}}(\lambda)$ has two local minima near 400 and 570 nm, which are related to a comparatively strong absorption of light by blood in these ranges. The blood itself has two clearly pronounced maxima of $A_{\mathrm{b}}(\lambda)$ at about 400 and 570 nm, with the right maximum of the dependence $A_{\mathrm{b}}(\lambda)$ being greater than the left one, whereas, for the curve $\mu_{\mathrm{ab}}(\lambda)$, the reverse situation is observed. This is explained by the fact that the absorption index of melanin decreases with increasing λ. As a result, more yellow-red light pene-

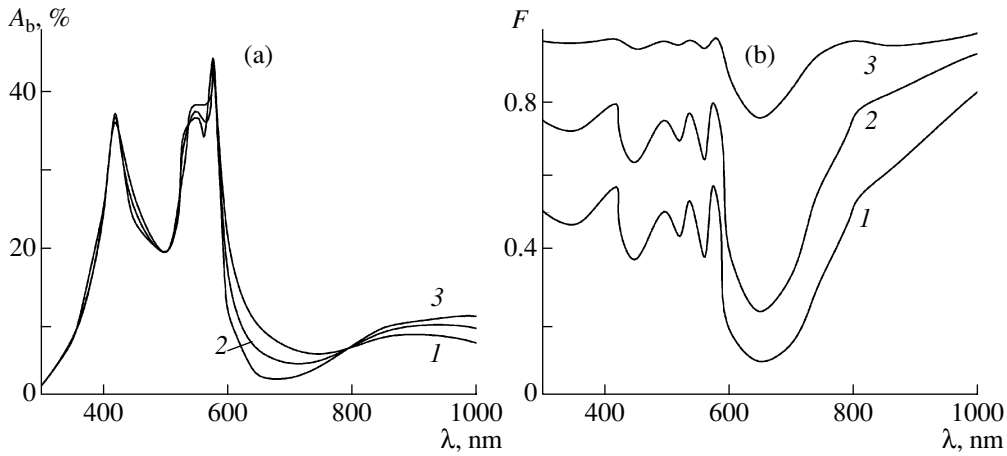

Fig. 2. Spectral dependences of the (a) blood absorption coefficient A_b and (b) oxyhemoglobin absorption fraction F at the oxygenation degree of blood S = (*1*) 0.5, (*2*) 0.75, and (*3*) 0.97.

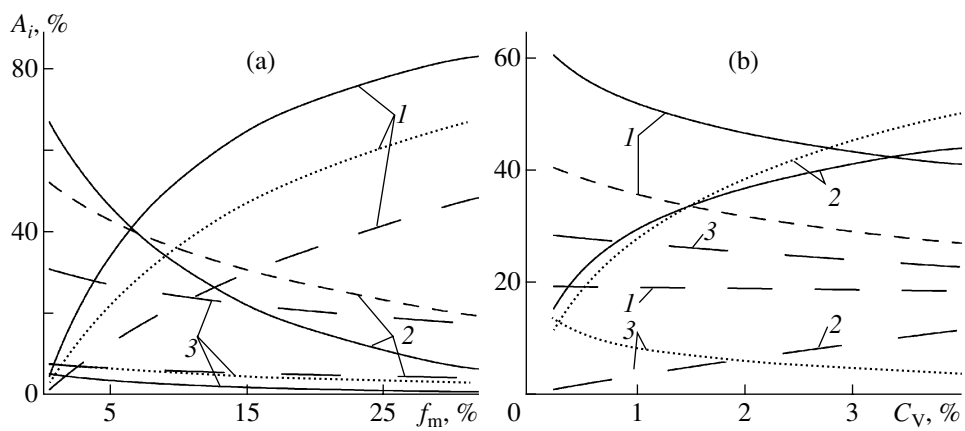

Fig. 3. Dependences of the absorption coefficient of (*1*) melanin, (*2*) blood, and (*3*) basic tissue (a) on the concentration of melanin f_m at $C_V = 2\%$ and (b) on C_V at $f_m = 8\%$ at the isosbestic points at (solid curves) 422, (dashed curves) 584, and (long-dashed curves) 800 nm.

trates into the dermis; therefore, blood integrally absorbs more strongly. For the basic tissue, $\mu_{at}(\lambda)$ monotonically decreases with λ, whereas the dependence $A_t(\lambda)$ has several extrema below 600 nm and monotonically increases at $\lambda > 600$ nm. These points of extrema occur due to the absorption of light by other components. For example, in the range where blood absorbs stronger, the basic tissue absorbs more weakly. An increase in $A_t(\lambda)$ in the red–near-IR spectral range has a similar explanation. Here, on the whole, $A_m(\lambda)$ and $A_b(\lambda)$ decrease with increasing λ. The light penetrates deep into the medium, which leads to an increase in the absorption coefficient of the basic tissue. Figure 1b shows that, in the spectral range 300–550 nm, the absorption of melanin is the strongest, whereas the absorption of the basic tissue is the weakest. In the range 550–750 nm, different relations between the absorption coefficients of the components are possible. The considerable absorption at λ = 750–1000 nm is attributed to the basic tissue and, as the light wavelength increases, the contribution of the basic tissue to the absorption increases compared to other components of the skin cover. Naturally, the total absorption coefficient of skin is $A(\lambda) = A_m(\lambda) + A_b(\lambda) + A_t(\lambda)$. As follows from Fig. 1b (curves *4*), the largest values of $A(\lambda)$ lie in the range $\lambda < 650$ nm and, to some extent, carry information on the spectral structure of the absorption indices of the components of skin. In the range $\lambda > 650$ nm, the spectrum $A(\lambda)$ is nearly neutral.

Above, we dealt with typical, healthy, light skin. In reality, its properties obviously differ for different people. The characteristic range of C_V is 1–3%, while that of f_m is 5–15% [10, 17]. This scatter in the parameters

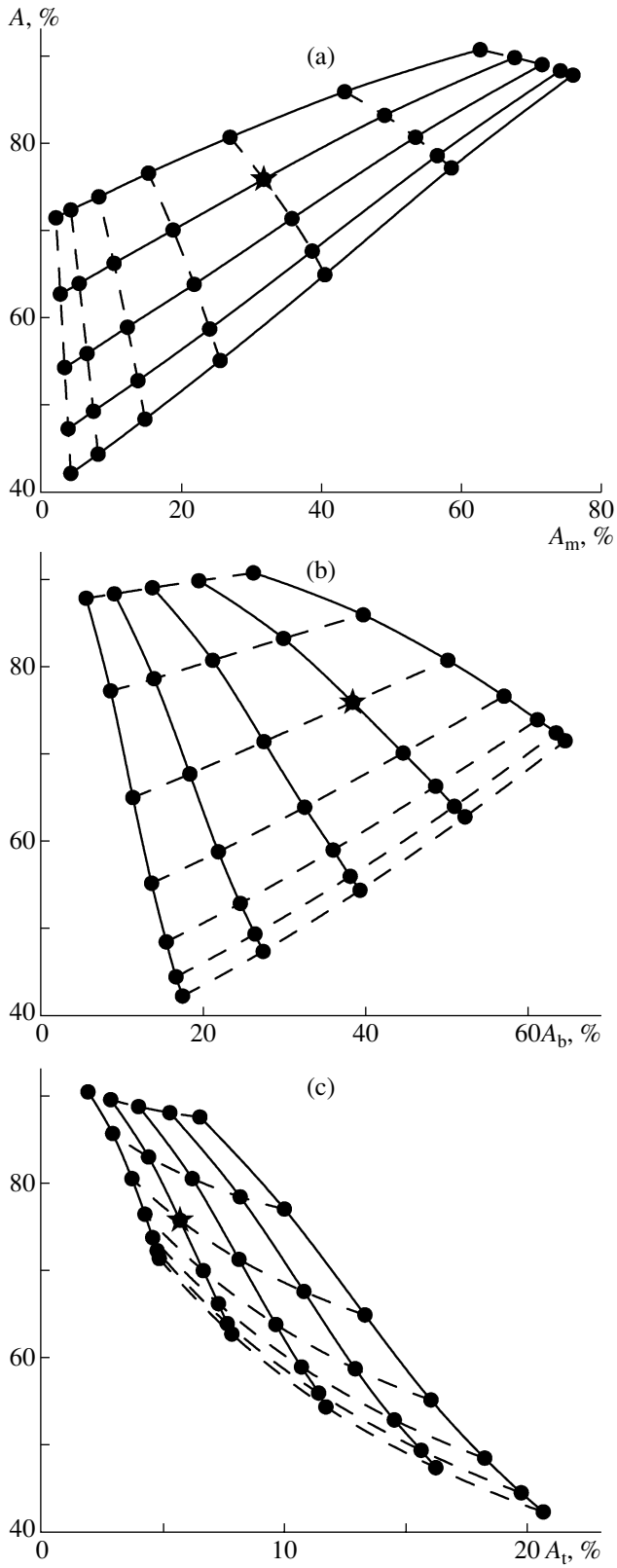

Fig. 4. Nomograms of the skin absorption coefficient with respect to the (a) melanin absorption coefficient, (b) blood absorption coefficient, and (c) basic tissue absorption coefficient at f_m = 0.5–32%, C_V = 0.5–4%, and λ = 584 nm.

determines the range of values of $A(\lambda)$, which is presented in Fig. 1b by two dashed curves *4*.

Figure 1b shows the spectrum of $A_b(\lambda)$ of the entire blood obtained at the oxygenation degree S = 0.75. Since the absorption indices of oxy- and deoxyhemoglobin are different, not only $A_{HbO_2}(\lambda)$ and $A_{Hb}(\lambda)$ but also $A_b(\lambda)$ will change with a change in S. Figure 2a presents the spectra of $A_{HbO_2}(\lambda)$ and $A_{Hb}(\lambda)$ at different values of the oxygenation degree of blood. It follows from Fig. 1a that, in the considered spectral range of 300–1000 nm, there are a number of isosbestic points at which the absorption indices of HbO_2 and Hb are equal to each other. Clearly, $A_{HbO_2}(\lambda)$ and $A_{Hb}(\lambda)$ coincide at these points. At other λ, there are differences, but they are small. Therefore, $A_b(\lambda)$ depends weakly on S. To analyze photoionization processes of HbO_2, it is interesting to estimate the ratio of the absorption coefficients of oxy- and deoxyhemoglobin over the spectrum at different values of the oxygenation degree. These data are presented in Fig. 2b. Here is given, the spectral dependence of the ratio $F(\lambda) = A_{HbO_2}(\lambda)/[A_{HbO_2}(\lambda) + A_{Hb}(\lambda)]$. It is seen that there is a characteristic range approximately between 600 and 800 nm where the fraction F of the light absorption by oxyhemoglobin in the total absorption of blood is considerably smaller than at other wavelengths. This is related to the specifics of the spectral indices $\mu_{a,\,HbO_2}(\lambda)$ and $\mu_{a,\,Hb}(\lambda)$ in this range.

Let us sequentially consider how the volume concentrations of melanin f_m and blood vessels C_V affect the absorption coefficients A_m, A_b and A_t at three isosbestic points located at 422, 584 and 800 nm. Figure 3a shows the dependences of these coefficients on f_m at C_V = 2%. It is seen that, with increasing f_m, A_m also increases. As λ increases, A_m decreases, since, in this case, the values of μ_{am} are smaller. At low concentrations f_m, the absorption coefficient $A_b > A_m$; however, with increasing f_m, due to the strong absorption of light in the epidermis, a smaller amount of light reaches the dermis and A_b decreases. As a result of competing absorption processes between melanin and hemoglobin at different f_m, as the wavelength increases, A_b can decrease or increase. This competition is clearly illustrated by the results presented in Fig. 3a. For example, upon passage from the light wavelength 422 nm to 584 nm, where f_m is small, and, therefore, the epidermis weakly absorbs light, the absorption coefficient A_b decreases because μ_{ab} decreases. At the same time, if the concentration f_m is large, the light at λ = 422 nm is nearly completely absorbed in the epidermis and only a small amount of the radiation reaches the dermis. At λ = 584 nm, more light penetrates the dermis, and A_b increases, though $\mu_{ab}(\lambda = 584\text{ nm}) < \mu_{ab}(\lambda = 422\text{ nm})$. Figure 3a also shows that A_t slowly decreases with an increasing concentration of melanin. Although the

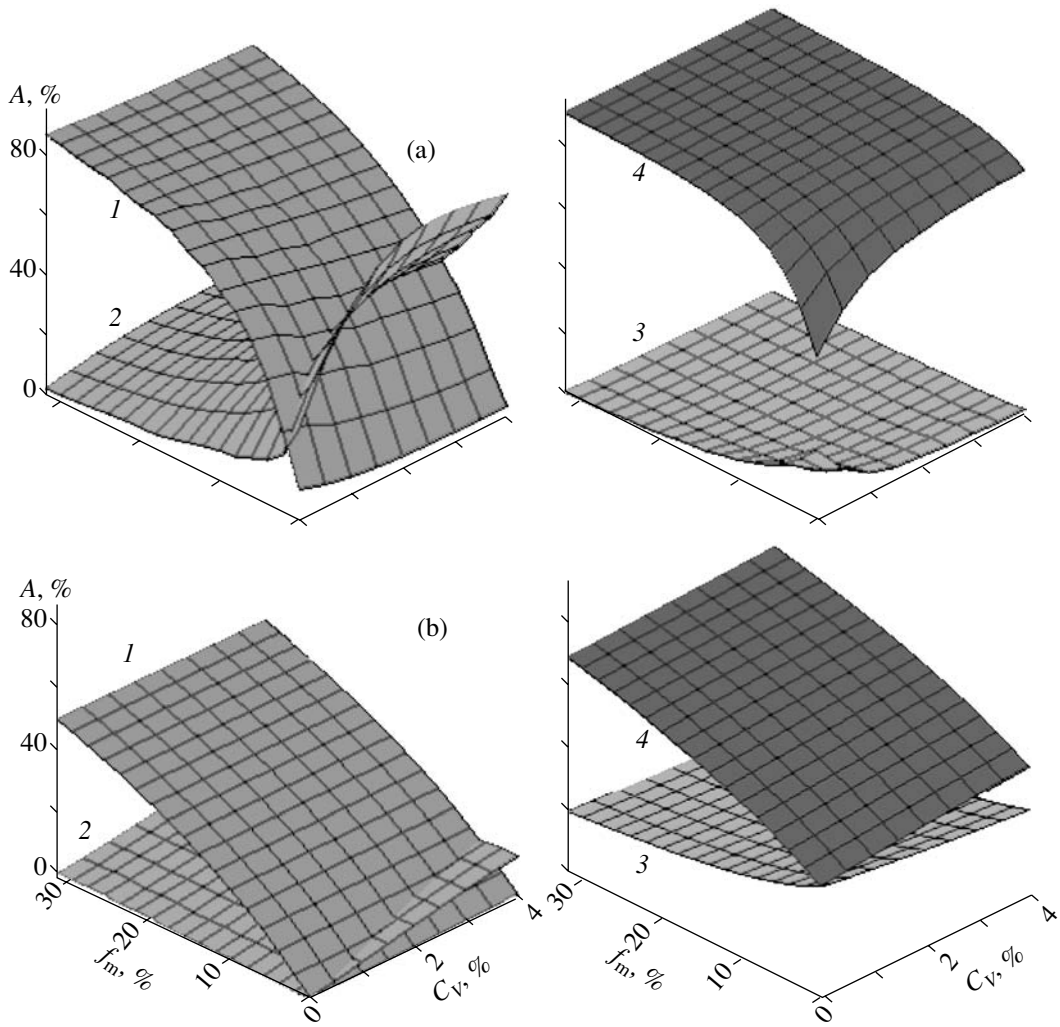

Fig. 5. Dependences of the absorption coefficient of (*1*) melanin, (*2*) blood, (*3*) basic tissue, and (*4*) entire skin cover on the concentrations f_m and C_V at the isosbestic points at λ = (a) 422 and (b) 800 nm.

index μ_{at} decreases with increasing λ, the absorption coefficient A_t increases strongly at λ = 800 nm, since the penetration depth of light into the dermis in the long-wavelength spectral range sharply increases [10].

Figure 3b presents the dependences of the absorption coefficients of the components on C_V at f_m = 8%. It is seen that, as λ increases, A_m decreases due to a decrease in μ_{am}. The absorption coefficient A_m also decreases when increasing the volume concentration of blood vessels. This is explained as follows. The light power absorbed in the epidermis is proportional to its radiation density $E(z)$, which, in turn, is determined by light fluxes that propagate in the direction of irradiation and in the opposite direction. The latter flux, which emerges from the dermis, decreases with increasing C_V. The competition between two absorption processes upon passage from the wavelength 422 nm to 584 nm is related to a decrease in μ_{ab} and to an increase in the amount of light emitted from the epidermis to the dermis. As a result of this competition, solid and dashed curves *2* intersect. Although the absorption of the blood at λ = 800 nm increases with increasing C_V, it remains insignificant because μ_{ab} is small in this spectral range. The absorption of light by melanin in this range is also small. Therefore, in the near-IR range, the absorption coefficient A_t of the basic tissue is the largest compared to other components of the skin.

In practice, only the light absorption coefficient A of the entire skin tissue can be measured. Clearly, only the diffusive reflection coefficient R of the skin should be determined for this purpose (see (4)). The relation between the absorption coefficient A and coefficients A_i of different components of the tissue can be found based on the modeling results obtained in this study. Taking λ = 584 nm as an example, Fig. 4 presents the data with the help of which all the sought absorption coefficients can be found if the concentrations f_m and C_V are known. In other words, the nomograms pre-

Absorption coefficients of normal skin and its basic chromophores

λ, nm	422	584	800
A, %	86 ± 1.5	75 ± 5	51 ± 1
Melanin, A_m, %	43 ± 8	31 ± 10	20 ± 2
Blood, A_b, %	40 ± 10	39 ± 11	6 ± 2.5
Basic tissue, A_t, %	3 ± 0.5	5 ± 1	25 ± 1

sented in Fig. 4 allow one to estimate the range of variation of the coefficients A_i for the known spectral value of A. These nomograms were obtained by performing an exhaustive search of almost all possible concentrations f_m and C_V for the skin, f_m = 0.5–32% and C_V = 0.25–4% [6, 10, 17]. Figures 4a, 4b, and 4c show the data for A_m, A_b, and A_t. In each of these figures, the coefficient A is plotted on the ordinate axis, while the values of A_i are plotted as abscissas. The data are presented for discrete values of f_m and C_V. Each solid curve corresponds to a constant concentration C_V at f_m = 0.5, 1, 2, 4, 8, 16, and 32%. Each dashed curve corresponds to a constant value of f_m at C_V = 0.25, 0.5, 1, 2, and 4%. The values of f_m and C_V successively increase along the corresponding solid and dashed curves. The asterisk indicates a point that corresponds to the average concentrations f_m and C_V for humans with light skin. These results show what maximal and minimal absorption coefficients A of the entire skin and of its individual components are possible, what the reasons for their occurrence are, and what the range of variation of A_i for each particular measured value of the coefficient $A = 1 - R$ is possible. Here, the data are presented only for the wavelength λ = 584 nm; however, they can be obtained for the entire spectrum in a similar way. Figure 5 shows, in a three-dimensional coordinate system, the surfaces that illustrate the dependences of the absorption coefficients A_m, A_b, A_t, and A of the entire skin and its components, respectively, on the concentrations f_m and C_V for other two wavelengths λ = 422 and 800 nm near the isosbestic points of the absorption indices $\mu_{a,\,HbO_2}$ and $\mu_{a,\,Hb}$. Similar to Fig. 4, these data make it possible to estimate the coefficients A, A_m, A_b, and A_t in wide ranges of variation of f_m and C_V, which cover actual values of these concentrations for normal and pathologically changed skin [10].

It was noted above that only the absorption coefficient of the entire skin can be experimentally determined rather than the coefficients of its individual components. The question arises as to what the range of scatter in A_i at a particular value of A is. The data of the table answer this question. Here, the second row gives the values of A at the wavelengths of the three isosbestic points for light human skin (f_m = 8%, C_V = 2%) and possible deviations from A for the typical ranges of f_m (5–15%) and C_V (1–3%) [10]. The remaining rows show the coefficients A_i and possible deviations from them. It can be seen from this table that, in the IR spectral range, where the absorption of melanin and blood is small, one can obtain rather exact information on the distribution of the absorption of light among the components of skin. In the visible range, there is a considerable scatter in the data on the absorption coefficients. Therefore, only estimative information on the relation between A_i and A can be obtained.

CONCLUSIONS

We mathematically studied the spectral behavior of the absorption coefficients of the entire skin and its chromophores, such as melanin, blood, oxy- and deoxyhemoglobin, and basic tissue. Since the direct data on A_m, A_b and A_t are experimentally unavailable, the method proposed seems to be reasonable alternative for the differentiation of the total coefficient A into its components (5)–(7). We showed that the spectral dependences $A_m(\lambda)$, $A_b(\lambda)$, and $A_t(\lambda)$ considerably differ from the corresponding dependences of the indices $\mu_{am}(\lambda)$, $\mu_{ab}(\lambda)$, and $\mu_{at}(\lambda)$. The reason is the spectral selectivity of the optical properties of the skin cover and, as a consequence, the strong dependence of the penetration depth of light on the radiation wavelength. In connection with this, the absorbed energy can be redistributed over λ and, for example, the absorption of blood in the green range of the spectrum can exceed that in the blue range or minima in the dependences $A_m(\lambda)$ can be observed in the range of the absorption maxima of hemoglobins. The absorption index $\mu_{at}(\lambda)$ of the basic tissue is the smallest among the indices of the other skin components, and the integrated light absorption of this tissue in the near-IR range is the highest. Our results make it possible to estimate the efficiency of the light–oxygen process [3], which is one of the theoretically and experimentally confirmed mechanisms by which laser radiation acts on biological tissues. After the introduction of the optical characteristics of a sensitizer into the model, such a modified model can be used in estimations of the efficiency of photodynamic transformations widely used in therapy of oncological diseases. The presented spectral data are the tool with the help of which researchers and medical practitioners can also quantitatively describe other biophysical and biochemical processes that proceed in tissues of the human skin during the laser therapy as a result of the absorption of external radiation.

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